

Finite Coherent Overlap Vertices and Their Local Contact Limit

Conditional EFT Interpretation and the Renormalization Boundary

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Abstract

Local quantum field theory writes interactions as pointwise products, for example $\int_X \lambda \phi(x)^4 dx$. In perturbation theory these contact vertices generate coincidence limits, momentum-conservation delta functions, and ultraviolet singularities. Building on the preceding papers on Dirac deltas, coherent Green functions, gauge fixing, and measurement effects, this paper develops the contact-interaction member of the same projection-kernel program. The central claim is narrow: a point-local interaction vertex is the zero-width limit of a finite coherent overlap vertex.

The mathematical core is an approximate-identity theorem. Given a bounded smoothing kernel $K_\epsilon(x, y)$, define the smeared field $\phi_\epsilon(x) = \int_X K_\epsilon(x, y) \phi(y) dy$ and the finite overlap interaction

$$S_{\text{int}, \epsilon}[\phi] = \frac{\lambda}{4!} \int_X \phi_\epsilon(x)^4 dx.$$

For smooth fields, $S_{\text{int}, \epsilon} \rightarrow S_{\text{int}} = \frac{\lambda}{4!} \int_X \phi(x)^4 dx$ as $\epsilon \downarrow 0$. On a compact spectral model, replacing the identity by a finite spectral projector Π_Λ yields a finite vertex tensor and finite coincident covariance in the retained sector. In momentum space, heat-kernel smearing multiplies propagators and vertices by rapidly decaying form factors, turning point-contact singularities into finite coherent-overlap expressions before the sharp limit is taken.

The MTT interpretation is conditional. If a selected physical sector supplies such a finite overlap kernel and an intertwiner to external fields, the local vertex may be read as its sharp limit. At nonzero width the construction is generally a nonlocal effective interaction, not automatically a local ultraviolet completion. A local parent mediator or an equivalent localization theorem would be needed to change that classification. The paper proves kernel convergence, the general n -point sharp limit, and finite-cutoff statements. It does not infer all-loop renormalizability, ultraviolet completion, Lorentzian causality, unitarity, or gauge/BRST consistency from finite width alone.

Version 1 Revision Note

Supersedes The unversioned April 2026 manuscript with the same subject.

Reason The earlier text correctly proved an approximate-identity limit but could be read as promoting finite overlap directly to a physical MTT interaction.

Resolution This version classifies the finite-width vertex as a nonlocal EFT interaction unless a local parent or localization theorem is supplied, and separates the proved finite-cutoff statements from all-loop, causal, unitary, and gauge claims.

Retained result The sharp-limit theorem and finite spectral vertex tensors are unchanged.

Remaining boundary A selected external-field source map and physical compatibility certificates are still required.

1 Purpose and claim discipline

The preceding papers established four projection-kernel replacements:

$$\delta(x - y) \rightsquigarrow K_{\text{coh}}(x, y), \quad LG = \delta \rightsquigarrow LG_{\text{coh}} = K_{\text{coh}},$$

$$\delta(G[A]) \rightsquigarrow \mathcal{K}_\epsilon(G[A]), \quad |x\rangle\langle x| \rightsquigarrow E_x^\epsilon.$$

The present paper treats the next major occurrence of delta-like structure: contact interactions.

In ordinary local field theory, an interaction such as

$$S_{\text{int}}[\phi] = \frac{\lambda}{4!} \int_X \phi(x)^4 dx$$

is point-local. In perturbation theory this point-locality appears as coincidence limits, delta functions enforcing exact vertex bookkeeping, and ultraviolet divergences.

The central thesis is:

A contact interaction is the zero-width limit of a finite coherent overlap vertex.

1.1 Non-claims

This paper does not claim that renormalization is unnecessary, that all quantum field theories become finite by the same smearing rule, or that a complete Standard Model calculation is performed here. The claims are narrower.

Level	Claim
Proved	Smearred overlap vertices converge to local contact vertices on smooth fields.
Proved	Finite spectral projection yields finite-dimensional vertex tensors and finite coincident covariance in the retained sector.
Standard analytic fact	Heat-kernel or spectral damping improves ultraviolet behavior in loop integrals.
MTT interpretation	Local contact vertices are singular downstream shadows of finite coherent overlap.
Research program	Renormalization can be partly re-read as repair of over-sharp projection.

1.2 Physical admissibility caveat

A finite-width vertex is not automatically a physically acceptable replacement for a local interaction. Generic nonlocal smearing can spoil microcausality, reflection positivity, unitarity, Ward identities, or BRST consistency. The present paper therefore distinguishes two notions:

arbitrary regulator kernel versus admissible coherent projection kernel.

Only the second has the intended MTT status. An admissible kernel must be generated by the same coherent-sector projection, overlap geometry, and symmetry constraints that define the effective theory. In gauge theories, for example, finite-width vertices must preserve the relevant Ward or Slavnov–Taylor identities. In Lorentzian settings, they must also be compatible with causal propagation on the admissible slab.

Thus the paper proves convergence and finiteness statements for controlled kernels, but does not claim that arbitrary smearing produces a consistent QFT.

Remark 1.1 (EFT classification). At fixed nonzero width, V_ϵ couples fields at separated points. It is therefore a nonlocal EFT vertex unless it is obtained by integrating out a declared local mediator or a localization theorem supplies an equivalent local parent theory. Finiteness of a vertex tensor at fixed cutoff does not imply all-loop renormalizability or ultraviolet completion.

2 Local contact vertices and their hidden delta structure

A local quartic vertex can be written formally as

$$S_{\text{int}}[\phi] = \frac{\lambda}{4!} \int_X \phi(x)^4 dx.$$

Equivalently, it is the four-field functional

$$S_{\text{int}}[\phi] = \frac{\lambda}{4!} \int_{X^4} V_\delta(x_1, x_2, x_3, x_4) \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) dx_1 \cdots dx_4,$$

where the contact vertex distribution is

$$V_\delta(x_1, x_2, x_3, x_4) = \int_X \prod_{j=1}^4 \delta(x - x_j) dx.$$

Thus the local vertex already contains a product of selection kernels. It says that all four fields interact at exactly the same point. The MTT diagnostic asks: what finite coherent overlap has been idealized as this exact coincidence?

3 Finite coherent overlap vertices

Let $K_\epsilon(x, y)$ be a family of smooth kernels on a compact Riemannian manifold (X, g) . Assume K_ϵ is an approximate identity: for every smooth f ,

$$K_\epsilon f(x) := \int_X K_\epsilon(x, y) f(y) dy \longrightarrow f(x)$$

in $C^\infty(X)$ as $\epsilon \downarrow 0$. Examples include heat kernels $H_\epsilon = e^{-\epsilon \Delta}$ and spectral cutoffs approaching the identity.

Define the smeared field

$$\phi_\epsilon(x) = K_\epsilon \phi(x) = \int_X K_\epsilon(x, y) \phi(y) dy.$$

The finite coherent overlap vertex is

$$S_{\text{int}, \epsilon}[\phi] = \frac{\lambda}{4!} \int_X \phi_\epsilon(x)^4 dx.$$

Equivalently,

$$S_{\text{int}, \epsilon}[\phi] = \frac{\lambda}{4!} \int_{X^4} V_\epsilon(x_1, x_2, x_3, x_4) \prod_{j=1}^4 \phi(x_j) dx_1 \cdots dx_4,$$

where

$$V_\epsilon(x_1, x_2, x_3, x_4) = \int_X \prod_{j=1}^4 K_\epsilon(x, x_j) dx.$$

For finite ϵ , the vertex no longer forces exact coincidence. It couples fields whose support overlaps within the coherent resolution scale of K_ϵ .

4 Theorem: overlap vertices converge to contact vertices

Theorem 4.1 (Contact vertex as sharp-overlap limit). *Let X be compact and let K_ϵ be an approximate identity on $C^\infty(X)$. For every $\phi \in C^\infty(X)$,*

$$S_{\text{int},\epsilon}[\phi] \longrightarrow S_{\text{int}}[\phi] = \frac{\lambda}{4!} \int_X \phi(x)^4 dx$$

as $\epsilon \downarrow 0$.

Proof. Since $K_\epsilon \phi \rightarrow \phi$ in $C^\infty(X)$, in particular $K_\epsilon \phi \rightarrow \phi$ uniformly. Write $\phi_\epsilon = K_\epsilon \phi$. Then

$$|\phi_\epsilon^4 - \phi^4| = |\phi_\epsilon - \phi| |\phi_\epsilon^3 + \phi_\epsilon^2 \phi + \phi_\epsilon \phi^2 + \phi^3|.$$

Uniform convergence implies $\sup_X |\phi_\epsilon - \phi| \rightarrow 0$, while ϕ_ϵ and ϕ are uniformly bounded for sufficiently small ϵ . Hence $\phi_\epsilon^4 \rightarrow \phi^4$ uniformly. Since X has finite volume,

$$\int_X \phi_\epsilon(x)^4 dx \rightarrow \int_X \phi(x)^4 dx.$$

Multiplying by $\lambda/4!$ gives the result. $\square$

Corollary 4.2 (Contact distribution as vertex limit). *In the distributional sense on X^4 ,*

$$V_\epsilon(x_1, x_2, x_3, x_4) \rightarrow V_\delta(x_1, x_2, x_3, x_4) = \int_X \prod_{j=1}^4 \delta(x - x_j) dx.$$

Proof. For a test function of product form $F(x_1, x_2, x_3, x_4) = \prod_{j=1}^4 \phi_j(x_j)$, the pairing with V_ϵ is

$$\int_X \prod_{j=1}^4 (K_\epsilon \phi_j)(x) dx,$$

which converges to

$$\int_X \prod_{j=1}^4 \phi_j(x) dx,$$

the pairing with V_δ . Finite linear combinations of product test functions are dense in the usual test-function topology on compact products; continuity gives the general case. $\square$

5 General n -point overlap vertices

The quartic case is not special. For an n -field local vertex, define

$$V_\delta^{(n)}(x_1, \dots, x_n) = \int_X \prod_{j=1}^n \delta(x - x_j) dx,$$

so that the corresponding local interaction is

$$S_{\text{int}}^{(n)}[\phi] = \frac{\lambda_n}{n!} \int_{X^n} V_\delta^{(n)}(x_1, \dots, x_n) \prod_{j=1}^n \phi(x_j) dx_1 \cdots dx_n = \frac{\lambda_n}{n!} \int_X \phi(x)^n dx.$$

The finite overlap replacement is

$$V_\epsilon^{(n)}(x_1, \dots, x_n) = \int_X \prod_{j=1}^n K_\epsilon(x, x_j) dx,$$

and

$$S_{\text{int},\epsilon}^{(n)}[\phi] = \frac{\lambda_n}{n!} \int_X (K_\epsilon \phi)(x)^n dx.$$

Proposition 5.1 (General n -point sharp-overlap limit). *Let X be compact, let K_ϵ be an approximate identity on $C^\infty(X)$, and let $n \geq 2$. For every $\phi \in C^\infty(X)$,*

$$S_{\text{int},\epsilon}^{(n)}[\phi] \longrightarrow S_{\text{int}}^{(n)}[\phi] = \frac{\lambda_n}{n!} \int_X \phi(x)^n dx.$$

Moreover $V_\epsilon^{(n)} \rightarrow V_\delta^{(n)}$ in distributions on X^n .

Proof. The functional convergence follows from $K_\epsilon \phi \rightarrow \phi$ uniformly and the identity

$$a^n - b^n = (a - b) \sum_{r=0}^{n-1} a^{n-1-r} b^r.$$

Uniform boundedness of $K_\epsilon \phi$ for small ϵ gives uniform convergence of $(K_\epsilon \phi)^n$ to ϕ^n , and compactness of X permits integration.

For the distributional statement, pair $V_\epsilon^{(n)}$ with product test functions $\prod_{j=1}^n \phi_j(x_j)$. The pairing is

$$\int_X \prod_{j=1}^n (K_\epsilon \phi_j)(x) dx,$$

which converges to

$$\int_X \prod_{j=1}^n \phi_j(x) dx.$$

Density of finite sums of product test functions in the test-function topology on X^n extends the result. $\square$

This general form is the useful one for perturbation theory: every point-local n -leg vertex is the sharp limit of a finite coherent n -fold overlap.

6 Finite spectral projection and finite vertex tensors

A particularly clean model uses a finite spectral projector. Let

$$\Pi_\Lambda f = \sum_{\lambda_n \leq \Lambda} \langle \phi_n, f \rangle \phi_n$$

be the spectral projector for a nonnegative Laplace-type operator on compact X . Define

$$\phi_\Lambda = \Pi_\Lambda \phi.$$

The projected quartic interaction is

$$S_{\text{int},\Lambda}[\phi] = \frac{\lambda}{4!} \int_X (\Pi_\Lambda \phi)(x)^4 dx.$$

Expanding

$$\Pi_\Lambda \phi = \sum_{\lambda_a \leq \Lambda} \phi_a e_a(x)$$

gives

$$S_{\text{int},\Lambda} = \frac{\lambda}{4!} \sum_{a,b,c,d \in I_\Lambda} V_{abcd} \phi_a \phi_b \phi_c \phi_d,$$

where $I_\Lambda = \{a : \lambda_a \leq \Lambda\}$ and

$$V_{abcd} = \int_X e_a(x) e_b(x) e_c(x) e_d(x) dx.$$

For finite Λ , this is a finite-dimensional tensor. There are no coincident-point products of distributions inside the retained sector.

Proposition 6.1 (Finite coincident covariance in a finite coherent sector). *Let $C_\Lambda = \Pi_\Lambda L^{-1} \Pi_\Lambda$, where L is strictly positive on $L^2(X)$. Then C_Λ has smooth finite-rank kernel*

$$C_\Lambda(x, y) = \sum_{\lambda_n \leq \Lambda} \frac{e_n(x) e_n^*(y)}{\mu_n},$$

where $Le_n = \mu_n e_n$ in the diagonal case. In particular, $C_\Lambda(x, x) < \infty$ for every $x \in X$.

Proof. The sum contains finitely many smooth terms, each with finite coefficient $1/\mu_n$. Therefore $C_\Lambda(x, y)$ is smooth and finite-rank. Evaluating at $x = y$ gives a finite sum of finite numbers. $\square$

This proposition is the rigorous finite-cutoff version of the slogan:

coincident-point singularities arise only after the coherent cutoff is removed.

7 Momentum-space model on the flat torus

Let $X = \mathbb{T}^d$ and use Fourier modes

$$e_k(x) = \frac{1}{(2\pi)^{d/2}} e^{ik \cdot x}, \quad k \in \mathbb{Z}^d.$$

For $L_m = -\Delta + m^2$, the free covariance is

$$C(k) = \frac{1}{|k|^2 + m^2}.$$

A sharp spectral cutoff retains $|k| \leq \Lambda$. The projected covariance is

$$C_\Lambda(x, y) = \frac{1}{(2\pi)^d} \sum_{|k| \leq \Lambda} \frac{e^{ik \cdot (x-y)}}{|k|^2 + m^2}.$$

For finite Λ , $C_\Lambda(x, x)$ is finite:

$$C_\Lambda(x, x) = \frac{1}{(2\pi)^d} \sum_{|k| \leq \Lambda} \frac{1}{|k|^2 + m^2} < \infty.$$

As $\Lambda \rightarrow \infty$, this diagonal quantity diverges for $d \geq 2$, logarithmically in $d = 2$ and by power law in $d \geq 3$. This is the standard ultraviolet coincidence problem.

A heat-filtered version replaces the sharp cutoff by

$$C_\tau(k) = \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}.$$

Then

$$C_\tau(x, x) = \frac{1}{(2\pi)^d} \sum_{k \in \mathbb{Z}^d} \frac{e^{-\tau|k|^2}}{|k|^2 + m^2} < \infty$$

for every $\tau > 0$. The divergence reappears only as $\tau \downarrow 0$.

Thus the finite coherent kernel does not “solve” the local theory by magic. It displays exactly where the singularity enters: at the zero-width limit.

8 Loop integrals and over-sharp projection

In Euclidean momentum space, the one-loop tadpole in a local scalar theory contains the formal integral

$$I_{\text{local}} = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + m^2},$$

which diverges for $d \geq 2$. A heat-coherent replacement gives

$$I_\tau = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{e^{-\tau k^2}}{k^2 + m^2}.$$

For every $\tau > 0$, $I_\tau < \infty$. The local divergent integral is recovered as $\tau \downarrow 0$.

This illustrates the structural point:

UV divergence is often the cost of taking a coherent overlap width to zero.

The statement is intentionally limited. It does not replace the full renormalization program. It explains why renormalization is encountered precisely where local field theory forms products or loops at exact coincidence.

9 Regulator versus coherent projection kernel

Mathematically, the kernels introduced above resemble familiar regulators. A heat factor $e^{-\tau k^2}$, a spectral cutoff Π_Λ , or a smooth compactly supported kernel can all improve ultraviolet behavior. The MTT interpretation is not that regulators are new. Rather, it changes the status of the finite object.

Ordinary regulator use	MTT coherent-kernel reading
auxiliary computational device	finite admissible overlap structure
removed after renormalization	may encode physical coherence scale
not usually part of ontology	downstream shadow of projection
chosen for convenience	constrained by coherent-sector geometry

Therefore the claim is not simply “add a cutoff.” The claim is that in an MTT-compatible description the finite kernel is the prior object, and the local contact vertex is the idealized limit. Ordinary renormalization remains necessary when one insists on taking that local limit.

10 Renormalization as repair of over-sharp projection

From the present viewpoint, renormalization has a two-sided interpretation.

First, it is the standard and indispensable method for defining local QFT after singular limits have been taken. Nothing in this paper challenges the causal-local or renormalization-group constructions [1, 2].

Second, it can be structurally re-read as a repair mechanism:

$$\begin{aligned} \text{finite overlap vertex} &\longrightarrow \text{zero-width contact vertex} \\ &\longrightarrow \text{coincident singularities} \longrightarrow \text{renormalization} \end{aligned}$$

In MTT terms, the local theory has projected away the finite overlap structure and then must restore predictivity by absorbing the consequences into parameters, counterterms, or running couplings.

Thus:

Renormalization repairs the downstream theory after admissible overlap has been idealized as point contact.

This is not a new computational scheme by itself. It is a structural reinterpretation of why the scheme is needed. In particular, the existence of a finite kernel does not abolish renormalization in the sharp local limit. It identifies the finite object whose removal creates the singular problem that renormalization then controls.

11 MTT interpretation

The MTT reading is direct.

Standard QFT object	MTT reading
$\delta(x - y)$	singular identity kernel
$\phi(x)^4$	zero-width contact overlap
V_δ	exact coincidence selection among four fields
V_ϵ	finite coherent overlap vertex
finite Λ or $\tau > 0$	retained coherent sector / admissible resolution
coincident singularity	failure of zero-width contact idealization
counterterms/running	repair of over-sharp projection in the effective theory

The native MTT object is not the local vertex V_δ , but the finite overlap vertex V_{coh} . The local vertex is the singular downstream limit.

12 Scope of the physical claim

The mathematical statements above are deliberately weaker than the possible physical interpretation. What is proved is that contact vertices arise as sharp limits of finite overlap vertices and that finite spectral sectors remove coincident distributional products inside the retained sector. What is not proved here is that a particular experimentally correct quantum field theory is obtained by a chosen kernel.

The physical MTT claim is conditional:

If an effective local QFT is the sharp limit of an admissible coherent projection, then its contact vertices should be read as zero-width overlap limits.

This conditional form is important. It avoids the false inference that any smoothing kernel is physically meaningful. Only kernels tied to the admissibility, symmetry, and projection structure of the underlying theory qualify as coherent kernels in the MTT sense.

13 Relation to the previous papers

The present paper extends the sequence as follows.

Previous paper	Replacement
Delta foundation	$\delta(x - y) \rightsquigarrow K_{\text{coh}}(x, y)$
Green functions	$LG = \delta \rightsquigarrow LG_{\text{coh}} = K_{\text{coh}}$
Gauge fixing	$\delta(G[A]) \rightsquigarrow$ finite gauge tube
Measurement	$ x\rangle\langle x \rightsquigarrow$ finite POVM effect
This paper	$\phi(x)^4 \rightsquigarrow$ finite coherent overlap vertex

The common pattern is now visible:

singular local object = zero-width limit of a bounded admissible kernel.

14 Scope of proof

The proof content of this paper is deliberately modest.

Claim	Status
Smeared smooth vertices converge to contact vertices	proved
Finite spectral sectors have finite vertex tensors	proved
Finite spectral covariance has finite diagonal	proved
Heat damping makes tadpole-type integrals finite	standard analytic fact, illustrated
General n -point contact vertices arise as sharp overlap limits	proved for smooth fields
Arbitrary smearing preserves locality, unitarity, and gauge symmetry	not claimed
All QFT renormalization is solved by MTT kernels	not claimed
Finite contact width proves all-loop UV completion	not claimed
The Standard Model has been recalculated with these kernels	not claimed

15 Conclusion

The local quartic vertex is the zero-width limit of a finite overlap vertex. At finite spectral bandwidth the vertex tensor and coincident covariance are finite in the retained sector.

The MTT reading is conditional: fixed width defines a nonlocal EFT until locality, causal support, unitarity, and gauge/BRST compatibility are proved. Renormalization remains the standard sharp-limit analysis.

References

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